

Revenue Leakage & Recovery Analysis

Power BI Dashboarding Project Proposal

1. Executive Summary:

This project aims to develop an interactive Power BI dashboard for a subscription software company to show how much of the money it bills actually reaches the bank, and where the rest is lost. Discounts, failed card payments, refunds, and customers cutting back their plans each leak a little money; together they can add up to a large loss that nobody tracks. The dashboard will focus on 23 key performance indicators built around one headline number, the **Revenue Capture Rate**: out of all the money the company should have earned, how much did it actually collect?

2. Problem Statement:

Background:

Subscription companies usually judge their performance by monthly recurring revenue (MRR), the total amount customers have agreed to pay each month. But MRR shows what the company expects to receive, not what reaches the bank. Along the way, money slips out through five gaps:

- Customers pay less than the standard price, or keep discounts for too long.
- Card payments fail, and the money is never collected.
- Payments are refunded.
- Customers reduce seats or move to a cheaper plan, and some then leave.
- Marketing money is spent on customers who leave before paying back what it cost to win them.

This is a common, well-documented problem. MGI Research found that 42% of companies lose revenue this way, and EY puts the loss at 1% to 5% of profits (EBITDA). Recurly, a subscription billing company, estimates failed payments alone cost businesses about \$129 billion in 2025, and that 20% to 40% of lost subscribers left only because a payment failed. Paddle found that customers who join on a discount are worth over 30% less over their lifetime. Yet finance sees invoices, marketing sees acquisition cost, and support sees complaints. With no view of the full journey of the money, nobody can say which leak costs the most or which to fix first.

Objective:

Develop a Power BI dashboard to monitor and analyse how much of the expected revenue is collected, where the rest is lost, and which leak to fix first. The dashboard will answer six questions:

- Q1.** Out of the money the company should earn each month, how much actually reaches the bank, and where is the biggest loss?
- Q2.** Which marketing channels bring in customers who pay back what it cost to win them, and how quickly?
- Q3.** Do customers who got a discount keep paying once the discount ends?
- Q4.** How much money is lost from customers whose payment failed, even though they never chose to leave?
- Q5.** Do warning signs such as plan downgrades, fewer seats, unresolved complaints, or no assigned account manager appear before revenue is lost, and how early?
- Q6.** If the company could fix only one leak next quarter, which one would recover the most money?

Scope:

Initial focus on four leak points (acquisition, pricing, collection, and account value) and two warning signals (support tickets and account ownership), across all plans, countries, industries, and marketing channels, month by month. Product usage, costs, and profit margins (not in the data), currency conversion, and estimating future outcomes are out of scope.

3. Data Sources:

Primary Data: A simulated (sample) dataset of a B2B subscription software company, made of five CSV files linked by customer_id and subscription_id:

- **Customers (CU), 5,000 rows:** industry, country, company size, sign-up date, the marketing channel the customer came from, assigned account manager, and current status.
- **Subscription Plan History (PH), 8,974 rows:** plan name, number of seats (users), price per seat, monthly price, start and end dates, and the reason for each plan change.
- **Invoices (INV), 126,852 rows:** billing month, amount due, amount paid, payment status (Paid, Failed, Refunded, Pending), payment date, and discount.

- **Support Tickets (TK), 45,658 rows:** category, priority, date opened, date resolved, and customer satisfaction (CSAT) score.
- **Marketing Spend (MK), 288 rows:** monthly spend per channel, new customers won, and leads generated.

Secondary Data: Published industry benchmarks, used only as reference lines on the dashboard: GRR and NRR medians (SaaS Capital, Benchmarkit 2026), CAC payback median (Benchmarkit 2026), failed payment share of lost subscribers (Recurly), and discount impact on customer value (Paddle).

4. Methodology:

Data Integration:

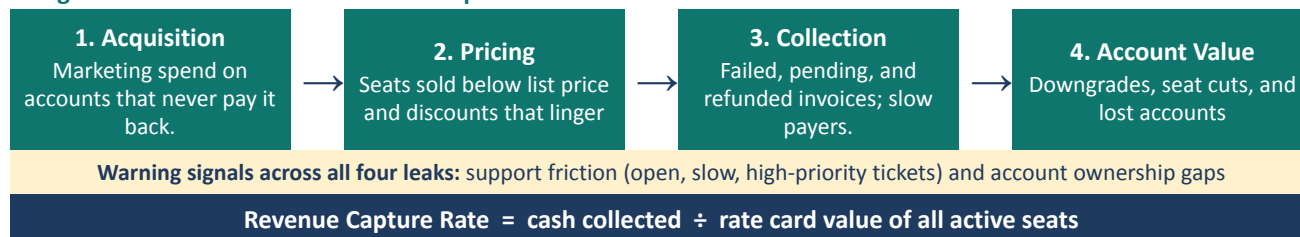
Extract the five CSV files into PostgreSQL, integrate them into one model, and load the prepared SQL views into Power BI:

- Load all five files into one relational model, profile every column in SQL and Power Query, and check key integrity: invoices or tickets with no matching customer, and plan rows with overlapping dates.
- Build a customer-by-month panel from valid_from and valid_to, so every account has exactly one active plan, or none, in every month.
- Derive the rate card: the standard price per seat for each plan, taken as the most common value per plan and period. Seats priced below it are flagged.
- **Match four numbers for every customer month:** rate card value, contracted value (monthly_price), billed (amount_due), and collected (amount_paid net of refunds). The gaps between them are the pricing and collection leaks.
- Trace every failed invoice forward to one of three outcomes (paid later, still pending, account lost) and record the days to recovery.
- Label each month-over-month change as New, Expansion, Contraction, Lost, or Reactivated. Group change_reason into a short reason list and label each lost account as a decided exit or a payment-led exit.
- Reconcile new_customers_acquired in the marketing file against actual signups by channel and month. Mismatches are reported, not forced to agree.
- Roll tickets up to customer month: count by priority, age of open tickets, resolution time, and CSAT where present.
- Missing values follow explicit rules; nothing is silently dropped:
 - CU: blank company size (10.2%) becomes an Unknown band. Blank account owner (35.5%) becomes Unassigned and is tested as a signal, not imputed.
 - PH: blank valid_to (55.7%) marks the current plan. Blank change_reason (1.7%) is kept as Unspecified. Blank price_per_seat (0.2%) is rebuilt from monthly_price ÷ seats.
 - INV: blank discount_pct (84.8%) means no discount. Blank payment_date (5.9%) equals the Failed (3.9%) plus Pending (2.0%) share, so blanks are unpaid invoices, confirmed row by row.
 - TK: blank resolved_date (9.1%) means open at export. CSAT (55.1% blank) is a directional signal only because responders are not a random sample.
 - MK: blank leads_generated (33.3%) limits conversion analysis to months with lead data. CAC and payback are unaffected.

Dashboard Design:

Key metrics are chosen by mapping each of the six questions to the leak it measures, so every visual answers one question.

Design framework: the Revenue Leak Map:



Every KPI belongs to one leak or to the warning layer; each leak is sized in currency so all four rank on one scale.

Key terms in simple words:

MRR	Money customers have agreed to pay every month	Seat	One user licence on a plan
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Rate card	The standard price per seat for each plan	CAC	Marketing money spent to win one new customer
Payback	Months until a customer's payments cover their CAC	Cohort	Customers who joined in the same month or through the same channel
GRR / NRR	Share of revenue kept from existing customers, without (GRR) and with (NRR) upgrades	Downgrade / Expansion	A customer moving to a cheaper or pricier plan, or to fewer or more seats
Silent exit	A customer lost because a payment failed, not by choice	P50 / P90	The middle value, and the value below which 90% of cases fall

Key Performance Indicators (KPIs):

23 KPIs grouped by leak point. **Shaded rows are the 14 advanced KPIs** (they join two or more tables, follow cohorts, or measure a window around an event). Data codes: CU customers, PH plan history, INV invoices, TK tickets, MK marketing spend.

KPI	How it is measured	Data	What it tells us
Headline			
Revenue Capture Rate	Cash collected in the month ÷ rate card value of all active seats (seats × list price per seat)	PH, INV	The one number the project is built around.
Leakage Mix	Each leak point's lost revenue ÷ total lost revenue in the period	All	Which leak to fix first.
Leak 1: Acquisition			
Channel CAC	Marketing spend ÷ new customers acquired, per channel and month	MK	Cost side of each channel.
Cash CAC Payback	Months until a channel cohort's average collected cash per account covers its CAC	MK, CU, INV	Uses banked cash, not billed revenue. B2B median about 16 months (Benchmarkit 2026).
12 Month Cash Multiple	Cash collected per account in its first 12 months ÷ CAC	MK, CU, INV	Observed return on spend with no lifetime assumptions.
Unrecovered Acquisition Spend	Sum of (CAC minus cash collected) for accounts lost before reaching payback	MK, CU, INV	Marketing money that never came back.
Leak 2: Pricing			
Rate Card Realization	Actual price per seat ÷ list price per seat for the plan, weighted by seats.	PH	Negotiated pricing drifting below list.
Discount Leakage	Discount value ÷ gross billed amount	INV	Size of the discount leak in currency.
Discount Tail Retention	Share of accounts still paying 3 cycles after their last discounted invoice, against matched full price accounts	INV, PH, CU	Whether discounts buy loyal accounts or only delay exits (Paddle research).
Leak 3: Collection			
Collection Efficiency	Amount paid net of refunds ÷ amount due	INV	Billed vs banked at invoice level.
Failed Payment Recovery Rate	Share of failed invoices after which the same account pays again within 2 cycles, plus median days to the next payment	INV	How well the business rescues payments that bounce.
Silent Exit Share	Lost accounts whose final invoice failed and who gave no cancellation reason ÷ all lost accounts, by count and by MRR	INV, PH, CU	Revenue lost from customers who never decided to leave. Industry range 20% to 40% (Recurly).
Collection Lag (P50, P90)	Days from start of billing month to payment date	INV	Slow payers surface here first.
Refund Ratio	Refunded amount ÷ collected amount, by plan and industry	INV	Cash that came in and went back out.
Leak 4: Account Value			
Revenue Bridge and Quick Ratio	Monthly New, Expansion, Contraction, Lost, and Reactivated MRR. Quick Ratio = (New + Expansion + Reactivated) ÷ (Contraction + Lost)	PH, INV	Growth efficiency. Above 4 is widely treated as healthy.
GRR and NRR	Revenue kept from a starting cohort, without and with expansion, trailing 12 months.	PH, INV	Private B2B medians: GRR 84% to 91%, NRR about 103% (SaaS Capital, Benchmarkit 2026).
Downgrade to Exit Rate	Share of accounts that downgrade and are lost within 6 months	PH, CU	Is a downgrade a warning or a stable resting point?
Tier Mobility Matrix	Monthly probability of moving between Starter, Growth, Pro, and Enterprise; median months to first upgrade	PH	The tier where accounts stall.

KPI	How it is measured	Data	What it tells us
Seat Compression Rate	Seats removed ÷ seats at start of month	PH	Quiet erosion before a plan change.
Warning Signals			
Friction Exposure	MRR held by accounts with open high-priority tickets older than the typical resolution time for that priority	TK, PH	Revenue sitting behind unresolved problems today.
Ticket Lift Before Downgrade	Ticket rate in the 90 days before a downgrade ÷ rate for matched stable accounts	TK, PH	How much warning support data gives.
Ticket Load per 10K MRR	Tickets ÷ (MRR ÷ 10,000), by plan	TK, PH	Which plans demand the most attention per unit of revenue.
Ownership Lift	NRR of owned minus unassigned accounts, compared within the same plan; share of MRR with no owner	CU, PH, INV	Puts a revenue value on the 35.5% of accounts nobody manages.

Analysis shown on the dashboard:

- **Leakage waterfall (Q1).** Monthly bridge from rate card value to collected cash, sliced by plan, industry, country, and channel.
- **Channel cohort cash curves (Q2).** Cumulative cash per account by months since signup, drawn against each channel's CAC line.
- **Matched cohort comparison (Q3).** Discounted vs. full-price accounts matched on plan and signup quarter in SQL. Power BI shows retention rate for both groups side by side with account count (n) labeled on each bar, so the size of the gap and the sample behind it are both visible on the chart.
- **Payment path analysis (Q4).** Trace every failed invoice to recovered, pending, or lost, shown as a flow diagram with recovery-lag distribution.
- **Lead and lag event study (Q5).** Tickets, open high-priority tickets, and seat cuts in the 30, 60, and 90 days before a downgrade or exit, built in SQL as a rolling window per account. Power BI shows the average for the downgrade group against the matched stable group, plus the lift ratio between them, the same measure as the Ticket Lift Before Downgrade KPI already in the KPI table.
- **Tier transition matrix (Q5).** Plan-to-plan transition probabilities to show where accounts stall or fall.
- **Segment heatmaps.** Revenue Capture Rate by industry × plan and country × channel to find where leaks cluster.
- **Recovery sizing (Q6).** What-if parameters, for example, raising failed payment recovery by 10 points or lifting unassigned accounts to the owned NRR, to size each fix in currency.

Dashboard pages:

A Power BI report with an EDA section (distributions through binning, percentile tables, scatter plots, null profiles) and a six-page dashboard: Leak Map, Acquisition Return, Price and Discount, Collections, Account Value, and Recovery Planner.

Interactivity:

Implement interactive features for drill-down analysis and trend exploration:

- Slicers for month, plan, industry, country, and channel; drill down in the leakage waterfall from month to plan to account; drill through from any leak to the accounts behind it.
- Tooltips that explain each KPI in plain words, and what-if sliders for recovery sizing and gross margin.

5. Expected Outcomes:

- An interactive dashboard, usable without technical help, showing the Revenue Capture Rate every month: how much of the expected money was actually collected.
- A clear picture of where the rest was lost: lower prices, discounts, failed payments, refunds, downgrades, or marketing spend that never paid back.
- A ranked list of leaks by how much money each one costs, so the company knows what to fix first.
- Marketing channels ranked by how fast their customers pay back their cost, not by customer count.
- Clear, side-by-side answers on whether discounts keep customers or only delay their exit, and how much money is lost from customers who left only because a payment failed.
- Early warning signs that appear before revenue is lost, and how much time each one gives the company to act.

6. Tools and Technologies:

- **Power BI** for dashboard development, with Power Query for data checks and exploratory charts.
- **SQL (PostgreSQL)** for data extraction and transformation. Anything that depends on order or time windows, such as what happened in the 90 days before a downgrade, is built here as views.
- **DAX** for KPI measures, ratios, and period-over-period comparisons. DAX only aggregates; sequence logic stays in SQL.

7. Risks and Challenges:

- **Integration challenges:** subscription_id may change with every plan change, and channel names may not match between the customer and marketing files. Both are checked in SQL before any KPI is built.
- **Data accuracy and consistency:** the dataset is simulated and has no official price list, so the standard price comes from the most common price per plan, and whether invoices are before or after discount is confirmed first. Every finding is checked against raw numbers, and any group comparison is shown with its account count so a gap built on very few accounts isn't presented as a stable pattern.
- **No cost or profit data:** payback is measured on revenue, not profit. A profit margin slider shows how results change under different assumptions, without inventing data.
- **Ease of use for non-technical users:** 23 KPIs could overwhelm people. The Revenue Capture Rate stays the headline, and tooltips explain every KPI in plain words.

8. Conclusion:

This project will give the company one clear, interactive view of its revenue: how much it should have earned, how much it actually received, and where the rest went. By following the money from marketing spend to the bank and putting a monetary value on each leak, the dashboard helps finance, marketing, and support teams agree on where to act first and back that decision with data.